

KULDEEP SINGH

EV Powertrain Validation & Controls Lead | EDU, Battery & Software

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PROFESSIONAL SUMMARY

EV powertrain software and validation leader targeting Electric Drive Unit validation, powertrain testing, vehicle controls, battery integration, and software-enabled test/data workflow roles. Brings 8 years in EV and hybrid commercial platforms with MATLAB/Simulink/Stateflow controls, BMS/eVCU software, HV battery architecture, DCDC/OBC/PDU integration, CAN diagnostics, test planning, DFMEA/RCA, Six Sigma, and ICAT/ARAI homologation. Strong fit for roles requiring powertrain testing at component, subsystem, and system levels plus cross-functional issue resolution.

TARGET ROLE FIT

- EV Powertrain Validation Engineer
- Electric Drive Unit / EDU Validation Engineer
- Vehicle Controls & Systems Integration Engineer
- Battery / Powertrain Software Validation Lead

CORE STRENGTHS

- Powertrain validation fit: delivered EV and hybrid platforms from architecture to vehicle validation, certification sign-off support, and issue closure using structured problem-solving tools.
- MATLAB/Simulink and data workflow fit: built range estimation models, control strategies, and verification artefacts; comfortable using model/data analysis to support efficient validation decisions.
- EDU/electrification fit: integrated motor, gearbox, HV battery, DCDC, OBC, PDU, charger, BMS, controller, and vehicle controls across commercial EV platforms.

PROFESSIONAL EXPERIENCE

Technical Manager - Product Development

IX Energy Pvt. Ltd., Noida, India | Jul 2018 - Present

- Led EV and hybrid system integration across battery, motor, gearbox, DCDC, OBC, PDU, charger, controllers, sensors, telematics, and vehicle control interfaces.
- Developed eVCU/BMS application software and operating-state logic using MATLAB/Simulink/Stateflow for drive readiness, charging mode, fault handling, power sequencing, and safe shutdown behaviour.
- Supported vehicle-level validation, test planning, fault investigation, CAN diagnostics, calibration behaviour review, supplier issue closure, and homologation evidence preparation.
- Applied DFMEA, RCA, DMAIC, Six Sigma, and quality tools to resolve integration issues, reduce validation cycles, and improve design robustness.
- Managed EV programme deliverables across internal engineering teams and external suppliers, communicating technical risk and readiness at system, subsystem, and component levels.

SELECTED PROGRAMMES & PROOF

- LCV/LPT EV conversion: ARAI-homologated 6.5T EV with 80 kW powertrain, 53 kWh / 320V LFP battery, and 119 km range; contributed ECU logic, charging interfaces, HV battery coordination, diagnostics, and validation support.
- 5T commercial EV: supported 80 kW EV conversion with 56 kWh / 310V NMC pack, power electronics interfaces, vehicle validation, and supplier coordination.
- P4 hybrid bus: integrated super-capacitor electrification kit for a 16T platform, supported ICAT validation, and contributed to 25% fuel-efficiency improvement evidence.

TECHNICAL TOOLKIT & CERTIFICATIONS

MATLAB, Simulink, Stateflow, powertrain validation, Electric Drive Unit, EDU, HV battery, BMS, eVCU, CAN, Vector CANoe, Peak CAN, test planning, instrumentation, DFMEA, RCA, Six Sigma, vehicle integration, ISO 26262, ICAT, ARAI.

- Six Sigma Black Belt - Certified
- ISO 26262 Functional Safety for Road Vehicles - TUV SUD Certified
- IATF 16949:2016 Automotive Quality Management - TUV SUD Certified
- AWS IoT & SIMULINK - Amazon Web Services / MathWorks Certified

EDUCATION

B.Tech - Mechanical Engineering, VIT University, Vellore, Tamil Nadu | 2014 - 2018 | First Class, 75%